Open the evidence document, **evidence.doc**

Make sure that your name, centre number and candidate number will appear on every page of this document. This document must contain your answers to each question.

Save this evidence document in your work area as:

evidence_ followed by your centre number_candidate number, for example: **evidence_zz999_9999**

A class declaration can be used to declare a record.

If the programming language does not support arrays, a list can be used instead.

A source file is used to answer **Question 2**. The file is called **`Characters.txt`**

1 A computer program is needed to store jobs in order of priority. Each job has a job number (for example, 123) and a priority from 1 to 10, with 1 being the highest priority and 10 the lowest.

The program stores the jobs in a global 2D array.

The pseudocode declaration for the array is:

```
DECLARE Jobs : ARRAY[0:99, 0:1] OF INTEGER
```

For example:

- `Jobs[0, 0]` stores the job number of the first job.
- `Jobs[0, 1]` stores the priority of the first job.

The global variable, `NumberOfJobs`, stores the number of jobs currently in the array.

(a) Write program code to declare the global 2D array `Jobs` and the global variable `NumberOfJobs`.

Save your program as **Question1_N22**. Copy and paste the program code into **part 1(a)** in the evidence document.

[3]

(b) The procedure `Initialise()` stores `-1` in each of the array elements and assigns `0` to `NumberOfJobs`.

Write program code for the procedure `Initialise()`.

Save your program. Copy and paste the program code into **part 1(b)** in the evidence document.

[3]

(c) The procedure `AddJob()`:

- takes a job number and priority as parameters
- stores the job in the next free array element
- outputs 'Added' if the job was successfully stored in the array
- outputs 'Not added' if the job was not successfully stored in the array.

Write program code for the procedure `AddJob()`.

Save your program.

Copy and paste the program code into **part 1(c)** in the evidence document.

[5]

(d) The main program should call the procedure `Initialise()` and then use the `AddJob()` procedure to store the following jobs in the order given:

Job number	Priority
12	10
526	9
33	8
12	9
78	1

Write program code for the main program and perform the tasks described.

Save your program.

Copy and paste the program code into **part 1(d)** in the evidence document.

[2]

(e) When a new job has been added, the array is sorted into ascending numerical order of priority using an insertion sort.

Write program code for the procedure `InsertionSort()` to sort the data into ascending numerical order of priority.

Save your program.

Copy and paste the program code into **part 1(e)** in the evidence document.

[5]

(f) The procedure `PrintArray()` outputs each job number and priority on a line, for example:

```
123 priority 1

39 priority 3

120 priority 7
```

Write program code for the procedure `PrintArray()`.

Save your program. Copy and paste the program code into **part 1(f)** in the evidence document.

[3]

(g) The main program needs to sort the array and then output the contents of the array.

(i) Amend the main program by writing program code to call procedures `InsertionSort()` and `PrintArray()`.

Save your program. Copy and paste the program code into **part 1(g)(i)** in the evidence document.

[1]

(ii) Test your program.

Take a screenshot of the output.

Copy and paste the screenshot into **part 1(g)(ii)** in the evidence document.

[1]

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